10th International Artificial Intelligence and Data Processing Symposium (IDAP'26), Sept 5-6, 2026, Türkiye (Malatya – İstanbul), Philippines

Depressive Belief Rigidity in Language Models: Representation and Limits of Activation-Based Control

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***Abstract*—Large-language models (LLMs) struggle with balanced cognitive restructuring in depressive contexts, a core component of effective depression therapy. In this work, we show that belief rigidity is linearly encoded in LLM internal representations but cannot be guided using contrastive activation addition steering to improve cognitive restructuring in depressive contexts. Specifically, we select and label real-world depressive texts for ordinal belief rigidity, and train linear probes on the last-layer token activations of the LLM. To steer the model, we generate matched pairs of high and low belief rigidity depressive texts for contrastive activation addition and evaluate steered therapist-style responses using an LLM-as-a-judge pipeline. Probes consistently outperform token count, sentiment, and TF-IDF models, suggesting that belief rigidity is encoded with moderate to strong strength in LLM internal activations, with strength peaking in mid to late layers. The steered model fails significantly to outperform baselines of wrong layer, random vector, and no intervention, indicating that LLMs possess a representation of belief rigidity, but the ability to perform psychotherapy via cognitive restructuring may be a more complex feature mediated by higher-order representations or reasoning circuits rather than by a simple rigidity axis in latent space.**

***Index Terms*—linear probes, activation steering, cognitive restructuring, belief rigidity, large language models**

I. Introduction

Large-language models (LLMs) have demonstrated impressive capabilities across a wide range of tasks [1]. Alongside the growing popularity of LLMs among the general public, these models have been increasingly used in sensitive therapeutic contexts, both in informal settings and in production systems [2]. Cognitive restructuring has emerged as one of the most widely accepted and effective methods in psychotherapy [3]. Cognitive restructuring involves identifying, challenging, and modifying irrational beliefs by shifting perspectives to enable patients to think more rationally and in a more balanced manner [4]. Despite the widespread adoption of LLMs for generating psychotherapeutic responses that involve cognitive restructuring, existing work suggests that these models struggle to produce high-quality outputs in complex real-world scenarios that require nuanced analysis of users' maladaptive beliefs [5]. In general, LLMs exhibit behaviors common in low-quality therapy, such as inaccurate assessments and a lack of reliability and safety [2]. Furthermore, existing techniques improving cognitive restructuring in LLMs rely on prompt-engineering—a method that lacks reliability and consistency [6], [7].

Linguistic patterns associated with depression often reflect rigid, absolute beliefs [8]. For instance, a depressive individual exhibiting such cognitive rigidity may describe their plight as *"impossible to overcome"* or assert that *"nothing will ever change"*, whereas a more flexible response may describe the situation as *"heading in the right direction"* or *"improving gradually with effort"*. Therapeutic interventions utilizing cognitive restructuring aim to cultivate belief flexibility in this manner. Central to the idea of belief rigidity is the capacity to model the beliefs of self and others, a faculty known as Theory of Mind (ToM) [9]. Recent literature has shown that ToM belief attribution is linearly encoded in LLM activations and can be modulated via activation steering to dramatically enhance ToM performance [10]–[12]. We hypothesize that belief rigidity is similarly encoded within the model's internal activations. Activation steering is an effective approach for optimizing model behavior along specific metrics or qualities [13]. This technique is predicated on the principle that if a feature is represented internally, direct intervention upon that representation should predictably alter the model's output.

A. Contribution.

In this work, we demonstrate that belief rigidity is linearly encoded in LLM activations and show that a linear probe outperforms baseline predictors based on TF-IDF, token count, and sentiment. Furthermore, we establish that LLMs cannot be steered using belief rigidity through Contrastive Activation Addition (CAA) to improve cognitive restructuring in psychotherapeutic contexts.

II. Related Work

A. Interpretability and Linear Probes

Prior work has leveraged linear probes to investigate internal representations in LLMs. Linear probes are simple, independently trained classifiers designed to diagnose features learned by deep neural networks [14]. Several works have demonstrated the utility of linear probes in assessing semantic and syntactic features of latent representations in natural language processing (NLP) models [15]–[18]. In addition to

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simple lexical features, linear probes have been employed to analyze latent belief representations in LLMs, particularly within the context of ToM. Existing research using probing suggests that LLMs can represent the beliefs of self and others [10], [11]. Additionally, ToM abilities in LLMs arise from emotional understanding rather than analytic reasoning [12]. These results all suggest that linear probes are effective at assessing latent features in LLMs and that LLM ToM capability is linearly encoded. Some work has proven especially promising for our hypothesis that maladaptive beliefs are also linearly represented given the similarity between ToM and belief rigidity [10]–[12]. Despite these advances, linear probes have yet to be applied to investigate belief rigidity—an important attribute of cognitive restructuring psychotherapy [4]. Our work aims to utilize these probes to investigate if depressive belief rigidity is linearly encoded in LLM activations and to identify the specific layers that encode it most effectively.

B. Activation Steering and Representation Engineering

Extant literature has examined the utility of activation steering for modulating LLM outputs. Contrastive Activation Addition (CAA) uses contrastive steering vectors to perturb model activations, demonstrating promise for mitigating hallucination, sycophancy, and refusal, among other behavioral attributes in LLM outputs [13], [19]. Furthermore, activation steering has been shown to enhance LLMs' ToM performance [10]–[12], suggesting its potential for understanding beliefs in psychotherapeutic contexts. Representation engineering through modifying and editing activations is also effective in modifying LLMs' ability to model complex psychological phenomena, such as depression [20]. While these approaches apply representation engineering to disparate tasks, we leverage them to steer belief rigidity toward improved cognitive restructuring in psychotherapy.

C. Behavioral Control and Alignment

Several approaches aim to improve LLMs' capacity for effective psychotherapy, cognitive restructuring, or cognitive reframing. These methodologies can be broadly categorized into fine-tuning, prompt engineering, system prompting, and multi-agent frameworks, often employing a hybrid of these techniques. Frameworks that rely solely on system prompting use a set of textual instructions and guidelines embedded within the LLM to steer outputs toward psychotherapeutic norms [21], [22]. Fine-tuning approaches specialize the LLM on curated datasets, often comprising ideal conversational responses [23]–[27]. For example, some work utilizes a combination of fine-tuning and system prompting, while others train multiple smaller LLMs for multi-turn supportive dialogues [23], [24]. In contrast, multi-agent systems comprise several LLMs that interact and converse to achieve the desired outcome [28], [29]. Despite the usefulness of these findings, prompt engineering is often susceptible to inconsistency and stability issues [30], [31]. Steering has also been shown to be more effective than fine-tuning or prompting in targeting specific behaviors [13], [19], [32].

III. Method

A. Task Definition.

Given an input query representing depressive or maladaptive beliefs x, the goal is to generate a psychotherapeutic response y that is consistent with the principles of cognitive-restructuring.

B. Setup

1) Model.: We use Mistral-7B-Instruct-v0.3 [33] with 4-bit quantization. We prefer this over the base model because Mistral-7B-Instruct-v0.3 is fine-tuned for task completion, enabling it to outperform the base model's text generation for psychotherapy. We use 4-bit quantization to enable efficient loading and inference on NVIDIA A100 GPUs.

2) Dataset.: We utilize the Dreaddit dataset [34], which consists of approximately 2,840 multi-domain social media posts for the purpose of stress identification as the source corpus for probing and steering experiments. We exclude posts from three subreddits: `food_pantry`, `assistance`, and `relationships`. These subreddits often contain logistics and support discussions rather than cognitively distorted, belief-driven content. From the filtered dataset, we sample a preliminary probing set of the first 1000 posts and steering set of the next 800 posts, choosing these sizes to balance statistical adequacy with the cost of LLM-based filtering. We then use GPT-4o [35] to further remove instances that do not exhibit depressive or maladaptive beliefs, as shown in Appendix E. This yields a final probing set of 775 posts and steering set of 604 posts used for all subsequent experiments, as shown in I. Our SQL queries are shown in Appendix H. This steering set is split further into a CAA vector computation subset (200 posts), a hyperparameter selection subset (30 posts), and an evaluation subset (374 posts). The 200-post CAA subset provides sufficient contrastive pairs to estimate stable mean activation differences without overfitting to idiosyncratic examples. The hyperparameter subset is kept small at 30 posts for computational efficiency, as grid searching over intervention layers and strengths requires repeated forward passes. The evaluation subset retains the remaining 374 posts to ensure final performance estimates are based on a sufficiently large and representative sample.

TABLE I
DATASET PARTITIONING AFTER FILTERING.

Split	n
Dreaddit (source)	$\sim$2840
Probing set	775
Steering set	604
CAA subset	200
Hyperparam subset	30
Evaluation subset	374

C. Probing Belief Rigidity

1) Dataset Labeling.: We use the GPT-4o pipeline to annotate the probing set with a belief rigidity score on an ordinal scale from 1 to 5, where 5 indicates flexible or minimally distorted beliefs, and 1 indicates highly rigid, absolutist, or cognitively distorted beliefs. This labeling enables the training of linear probes to capture the degree of belief rigidity in model representations. We show our LLM-as-judge labeling prompt in Appendix E.

2) Activation Extraction.: For each Dreaddit instance, we run the model in inference mode to extract the hidden-state activations of the final non-padding token across all layers. For an input sequence of length T, we denote the activation at layer $l \in \{1, \ldots, L\}$ as:

$$x_l = h_l^{(T)} \in \mathbb{R}^d, \tag{1}$$

where d represents the model's hidden dimension. This procedure yields a discrete residual stream vector per layer for each example, which subsequently serves as the primary feature set for linear probing.

a) Linear Probe.: For each layer $l \in \{1, \ldots, L\}$, we train a separate linear probe f_l to predict the ordinal belief rigidity $y_i \in \{1, \ldots, 5\}$ from the extracted activations $x_{i,l} \in \mathbb{R}^d$. The probe is optimized using the Ridge regression objective:

$$\min_{w_l, b_l} \sum_{i=1}^{N} (w_l^\top x_{i,l} + b_l - y_i)^2 + \alpha \|w_l\|_2^2 \tag{2}$$

where the regularization strength α is selected via the inner cross-validation loop described below.

The predicted rigidity $\hat{y}_{i,l}$ for a given activation is then computed as the continuous scalar output of the linear model:

$$\hat{y}_{i,l} = w_l^\top x_{i,l} + b_l \tag{3}$$

3) Evaluation.: We evaluate predictive correlation using Spearman's rank correlation coefficient

$$\rho_l = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}, \tag{4}$$

where $d_{i,l} = R(\hat{y}_{i,l}) - R(y_i)$ is the difference between layer l's ranks for each observation i, and n is the total number of observations. To quantify the average magnitude of error in our probe's predictions across n instances, we calculate the Mean Absolute Error (MAE) for each layer l's probe:

$$\text{MAE}_l = \frac{1}{n} \sum_{i=1}^{n} |\hat{y}_{i,l} - y_i|, \tag{5}$$

where $\hat{y}_{i,l}$ is the continuous predicted belief rigidity at layer l, and $y_i \in \{1, \ldots, 5\}$ is the corresponding belief rigidity label.

a) Nested Cross-Validation.: To obtain unbiased performance estimates and avoid optimistic reporting from post-hoc layer selection, we employ a nested k-fold cross-validation scheme with $k = 5$ outer folds and $k = 5$ inner folds. In the outer loop, each fold is held out in turn and never used for any selection decision. Within the outer training portion, an inner 5-fold CV selects both the optimal layer l^* and regularization strength α^* from the grid $\alpha \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$, using mean Spearman ρ as the selection criterion. A probe is then refit on the full outer training fold using (l^*, α^*) and scored once on the held-out outer fold. Final reported metrics are means and standard deviations across the 5 outer folds, with 95% bootstrap confidence intervals computed by pooling outer-fold predictions.

D. Validation and Baseline Comparisons

We compare the results of our linear probe against several other models to confirm that the probe is identifying a genuine latent representation of belief rigidity rather than high-dimensional noise or surface-level heuristics.

- We generate a null distribution of activations x_l, and randomized labels y_i. If the original probe significantly outperforms the probe trained on this randomized correspondence, it confirms the presence of an authentic signal.
- Depressive texts often exhibit a correlation between length and severity. We train a ridge regressor on the total non-padding token length (scalar) of each text T_i to confirm that the probe is not using length as a proxy for belief rigidity.
- We use a DistilBERT sentiment classifier model to predict belief rigidity from continuous valence scores s_i [36]. We train a ridge regressor on s_i to validate that our probe captures belief rigidity and not general negative sentiment.
- We train a ridge regressor on sparse TF-IDF vectors to test if belief rigidity is a function of rigid keyword frequencies (e.g., "always," "never") rather than the model's internal representations.

E. Contrastive Activation Addition

1) Steering Direction.: For each depressive Dreaddit instance, we instruct GPT-4o to reframe the text by generating a version with reduced belief rigidity. This re-framed version contains less rigid, less absolute, and more flexible beliefs. Our prompt can be found in Appendix F.

2) Intervention Mechanism.: To construct the steering vector, we compute the mean activation difference across each matched pair of high and low (GPT-4o reframe) belief rigidity Dreaddit instances. Specifically, for the chosen layer, we extract the last-token residual stream representation from the original prompts and the re-framed prompts, and calculate the mean difference:

$$d = \frac{1}{N} \sum_{i=1}^{N} \left(h_i^{\text{reframe}} - h_i^{\text{original}} \right),$$

normalized to unit norm. At inference, this steering vector is added to the model's residual stream output at the chosen layer, scaled by an intervention strength, and restricted to positions beyond the system instruction:

$$h \leftarrow h + \alpha \, m \odot d,$$

where m is a binary mask selecting post-instruction tokens and α denotes the intervention strength. The forward hook uses KV caching to efficiently generate token-by-token outputs without recomputing attention matrices.

3) Hyperparameter Selection.: We identify the optimal layer l^* and intervention strength α for final held-out evaluation by performing a localized grid search over the hyperparameter space. Specifically, we select the layer l_{best} associated with the highest-performing probe, along with its three adjacent layers $l \in \{l_{best}-1, l_{best}, l_{best}+1, l_{best}+2\}$. We evaluate four discrete intervention strengths: $\alpha \in \{0.5, 1, 2, 4\}$. Subsequently, we generate cognitive restructuring (CBT-style) outputs for each (l, α) combination using a dedicated hyperparameter selection subset ($n = 30$), and determine the optimal configuration via an LLM-as-a-judge framework for final evaluation.

F. Response Generation and Evaluation

1) Prompting Setup.: We evaluate the model's ability to generate psychotherapy responses consistent with cognitive restructuring by using the same prompt across all steering configurations. Our prompt reflects the three core components of cognitive restructuring: identifying maladaptive beliefs, examining evidence, and proposing alternatives to change behavior and reduce stress [3]. To maximize psychotherapeutic relevance, we ask the LLM to adopt a psychotherapist persona. We restrict model output to 150 tokens for computational efficiency. Our prompt for Mistral-7B psychotherapy generation is provided in Appendix G.

2) Experimental Conditions.: We establish four separate steering configurations. Each configuration's inputs consist of the same held-out final evaluation subset from Dreaddit ($n = 374$). We first perform inference using the **baseline model** with no intervention ($\alpha = 0$) as a control to validate the cognitive restructuring impact of CAA. We then perform inference using the **standard CAA configuration** with the optimal layer and α identified in the hyperparameter selection step. Next, we evaluate the **random-direction steered** model. This configuration uses the same intervention mechanism as the standard CAA configuration, except that the steering vector d is defined as $d \sim \mathcal{N}(0, I) \in \mathbb{R}^d$, where the vector components are sampled from a normal distribution and then normalized. This confirms that the standard CAA vector is steering the model's representations towards lower belief rigidity, rather than injecting random noise. Our final configuration is **wrong layer CAA**, where we add the steering vector at the residual stream output for a low-performing layer rather than the optimal layer, to test whether steering's efficacy depends on the localization of belief rigidity or whether any shift in the residual stream can induce these changes.

3) Evaluation.: We use an LLM-as-a-judge pipeline with the GPT-4o model to evaluate the model's hyperparameter selection outputs and final experimental configuration outputs on the evaluation subset. We prompted the model to grade responses on a 5-point ordinal rubric ranging from 0 to 4, based on the quality of cognitive restructuring, belief flexibility, avoidance of surface-level emotional validation, and overall helpfulness, where 0 indicates the poorest performance and 4 the best. Our LLM prompt is shown in Appendix D.

4) Statistical Analysis: The unsteered configuration ($\alpha = 0$) serves as the reference baseline. We evaluate six rubric metrics — *therapy_quality*, *identifies_belief*, *asks_evidence*, *avoids_validation*, *proposes_next_step*, and *belief_flexibility* — plus a composite omnibus score defined as their per-item mean. Uncertainty is quantified via percentile bootstrap (10,000 resamples), with items resampled jointly across all configurations using a shared index matrix to preserve within-item pairing; this yields 95% CIs on every mean and paired difference. Significance is assessed with a paired sign-flip permutation test (10,000 permutations, two-sided, Phipson–Smyth $+1$ correction), which makes no normality assumption on the ordinal 0–4 grades. Effect size is reported as Cohen's d_z (mean difference divided by SD of within-item differences). Multiple comparisons are controlled with Holm–Bonferroni correction applied within two planned families separately: 18 tests for the primary baseline-referenced comparisons (6 metrics $\times$ 3 configurations) and 6 tests for the specificity contrast comparing the real steering vector against a random vector at the same layer and α. Composite rows are reported uncorrected and labeled as such.

To address the concern that a non-significant p-value alone does not constitute evidence of no effect, we additionally conduct equivalence testing (TOST, $\alpha = .05$) with a smallest effect size of interest of ± 0.2 baseline SD. Equivalence is established when the 90% bootstrap CI of the paired difference falls strictly within this bound, inheriting the same distribution-free machinery as the primary tests. A robustness check reruns every comparison as a signed-rank permutation test, relaxing the interval-scale assumption on the Likert grades entirely; results are consistent throughout and reported in the supplement. Two limitations apply: the ± 0.2 SD equivalence bound was not pre-registered and represents a conventional small-effect threshold that reviewers may contest, and the use of a single LLM judge precludes separation of judge noise from the reported uncertainty intervals.

IV. Results

A. Probe Results

1) Layer-wise Performance. : As shown in Figure 1, results indicate that the belief rigidity probe consistently outperforms the shuffled label probe across all metrics and nearly all layers. Our probe achieves its highest Pearson correlation coefficient values (~ 0.6) at and after layer 15, indicating a moderate to positive linear relationship, while the shuffled-label probe hovers near zero with no systematic trend across layers, suggesting that there is no linear relationship between activations and belief label. Spearman's rank correlation coefficient results also reflect these trends, indicating a strong positive monotonic relationship. Finally, as indicated in Appendix A and Appendix B, our base probe achieves peak mean absolute error (MAE) values of $\sim 0.6 - 0.7$ after layer 15, while the

shuffled label probe exhibits MAE values of $\sim 1.0 - 1.1$. Interestingly, MAE decreases as we move to later layers for the base probe, but increases for later layers of the shuffled label probe. The consistent improvement in probe performance at deeper layers is consistent with the broader mechanistic interpretability literature on representational geometry. Early layers in transformer models are known to encode relatively low-level syntactic and positional features, while intermediate and later layers progressively build more abstract, semantically structured representations of the input [37], [38]. Belief rigidity, as a high-level psychological construct, is unlikely to be linearly decodable from early-layer activations precisely because those layers have not yet integrated sufficient contextual information to represent the patient's epistemic stance toward their belief. By layer 15, the residual stream has accumulated contributions from many attention heads and MLP sublayers, and prior work suggests that MLP layers in particular are responsible for storing and surfacing factual and conceptual knowledge [39]. The improvement in Pearson and Spearman correlations alongside the reduction in MAE at later layers therefore suggests that belief rigidity is encoded as an increasingly linear and decodable feature of the residual stream as the model builds a richer contextual representation of the conversation. The divergence between the base and shuffled-label probes in later layers further supports this interpretation: if later-layer activations contained no structured information about belief rigidity, the shuffled probe's MAE would not systematically worsen relative to the base probe, as there would be nothing meaningful for the base probe to exploit.

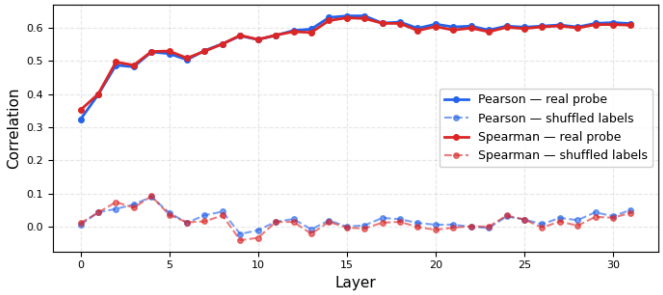


Fig. 1. Layer-wise belief rigidity probe results

2) Baseline Comparison.: Table II reports nested cross-validation performance for the belief rigidity probe and all baselines. The probe achieves a mean Spearman ρ of 0.671 ± 0.025 and MAE of 0.583 ± 0.033, substantially outperforming all baselines across every metric, with non-overlapping 95% bootstrap confidence intervals in each case. TF-IDF is the strongest baseline at $\rho = 0.467$, reflecting the contribution of rigid lexical markers such as absolutist words; the probe's advantage of roughly 0.20 Spearman points beyond TF-IDF indicates that LLM activations encode belief rigidity over and above what surface keyword frequencies capture. Sentiment performs modestly ($\rho = 0.219$), confirming that belief rigidity is a distinct construct from general negative valence. Token length is near zero and slightly negative ($\rho = -0.084$), ruling out post length as a confound. The narrow standard deviations across outer folds (≤ 0.057 for all models) indicate stable estimates. Taken together, these results provide evidence that belief rigidity is linearly encoded in the model's internal representations as a feature distinct from lexical, sentiment, and length-based signals. For comparison, the non-nested protocol—selecting the best-performing layer after inspecting all layers on the same cross-validation folds—yields a Spearman of 0.631 at layer 15, approximately 0.04 above the nested estimate, confirming that post-hoc layer selection introduces modest optimism in performance reporting.

TABLE II
NESTED CV PROBE VS. BASELINE PERFORMANCE
(MEAN ± SD, 95% BOOTSTRAP CI).

Model	Pearson	Spearman	MAE
TF-IDF	0.473 ±0.022 [0.411, 0.515]	0.467 ±0.019 [0.397, 0.517]	0.721 ±0.044 [0.687, 0.757]
Sentiment	0.198 ±0.057 [0.104, 0.251]	0.219 ±0.056 [0.054, 0.202]	0.835 ±0.054 [0.802, 0.870]
Token Len.	−0.063 ±0.034 [−0.157, −0.019]	−0.084 ±0.039 [−0.160, −0.021]	0.851 ±0.047 [0.820, 0.886]
Probe	**0.673** ±0.024 [0.627, 0.703]	**0.671** ±0.025 [0.619, 0.705]	**0.583** ±0.033 [0.552, 0.614]

B. Steering Results.

1) Grid Search.: As shown in Table III, we choose layer 17 with $\alpha = 1$ for steering, as this configuration exhibits the most consistent performance across the qualitative LLM-as-judge metrics in the grid search. Full results for all configurations are available upon request

TABLE III
BEST-PERFORMING STEERING CONFIGURATION.

Layer	α	TQ	IB	AE	AV	PNS
17	1	2.93	2.23	2.5	3.47	2.47

TQ: Therapy Quality; IB: Identifies Belief; AE: Asks for Evidence; AV: Avoids Validation; PNS: Proposes Next Step.

2) Final Evaluation.: Table IV reports rubric scores for the selected steering configuration (layer 17, $\alpha = 1$) against the unsteered baseline. No metric reaches significance after Holm–Bonferroni correction (all $p = 1.000$), and all paired differences are small in magnitude ($|d_z| \leq 0.07$). Critically, five of six individual metrics and the composite satisfy formal equivalence at the ± 0.2 baseline SD bound, meaning the data actively rule out effects larger than a conventional small threshold rather than merely failing to detect them. The one exception is *avoids_validation*, whose confidence interval marginally crosses the upper equivalence bound, leaving that metric inconclusive rather than equivalent. The specificity contrast — comparing the real steering vector against a random vector at the same layer and α — yields no significant differences across any metric (full results in Appendix C), indicating that the belief rigidity direction confers no advantage over arbitrary activation perturbation. Together, these results support the conclusion that CAA steering along the belief rigidity axis does not meaningfully improve psychotherapeutic cognitive restructuring, and that this null result is bounded rather than merely underpowered.

TABLE IV

CBT RUBRIC SCORES FOR THE SELECTED STEERING CONFIGURATION (L17, $\alpha = 1$) VERSUS THE NO-STEERING BASELINE, WITH PAIRED INFERENCE. Δ IS STEERED MINUS UNSTEERED. EQUIV. DENOTES EQUIVALENCE AT $\alpha = .05$ (± 0.2 BASELINE SD BOUND). p VALUES ARE HOLM–BONFERRONI CORRECTED.

Metric	Baseline	Steered	Δ [95% CI]	d_z	p_{Holm}	Equiv.
Therapy quality	2.764	2.764	+0.00 [−0.07, +0.07]	+0.01	1.000	✓
Identifies belief	2.338	2.373	+0.04 [−0.05, +0.11]	+0.04	1.000	✓
Asks for evidence	2.397	2.434	+0.04 [−0.06, +0.13]	+0.04	1.000	✓
Avoids validation	3.080	3.145	+0.06 [−0.02, +0.15]	+0.07	1.000	–
Proposes next step	2.239	2.268	+0.03 [−0.07, +0.13]	+0.03	1.000	✓
Belief flexibility	3.088	3.105	+0.02 [−0.05, +0.08]	+0.03	1.000	✓
Composite	2.651	2.681	+0.03 [−0.02, +0.08]	+0.06	–	✓

V. DISCUSSION

A. Belief Rigidity Representation

Our belief-rigidity probe consistently outperforms baselines (Table II), suggesting that it reflects a distinct semantic feature in the model's latent space. However, negligible improvements in cognitive restructuring ability (Tab. IV) suggest that the probed belief-rigidity representation may be decoupled from the ability to re-frame thoughts as a psychotherapist. We offer two possible explanations for these results.

- Belief rigidity may be a correlate or feature rather than the cause of cognitive restructuring ability. Cognitive restructuring may be driven by higher-level features such as logical flexibility, epistemic humility, or perspective shifting that are distinct from the singular axis of belief rigidity [40], [41].
- Our LLM-as-a-judge prompt explicitly instructs the judge to refrain from scoring linguistic style only and rather focus more on the quality of psychotherapeutic reasoning. Our belief rigidity steering direction may primarily capture semantic content, making it insufficient to substantially improve cognitive restructuring, which may rely on a distinct set of reasoning or task-oriented transformer circuits [42], [43].

B. Limitations

The findings of this study need to be considered in light of certain limitations. For steering, we run inference on only 374 samples and truncate outputs to 150 tokens; the former limits statistical robustness, while the latter may cause the model to cut off mid-response, potentially truncating critical cognitive restructuring outputs before the LLM-as-judge can evaluate them. Additionally, our LLM-as-judge pipeline uses only one model and grades outputs in batches of five per API call; while this improves cost-efficiency, it limits grading rigor and may introduce bias from neighboring outputs within the same batch. Finally, as experiments are conducted on a single LLM, our findings may not generalize beyond this model family.

VI. CONCLUSION

In this work, we use linear classifier probes to show that belief rigidity is linearly encoded in LLM activations, with moderate-to-strong correlation, and that this encoding significantly outperforms baseline predictors. We attempt to steer the model towards improved psychotherapeutic cognitive restructuring using contrastive activation addition with high and low-rigidity pairs, but find negligible improvements over baselines and no-intervention baselines. These results suggest that LLMs possess a representation of belief rigidity, but the ability to perform psychotherapy may be driven by higher-order features or more complex reasoning or logic circuits rather than a simple belief rigidity direction in latent space. Future work should investigate whether mechanistic controls can reliably prevent models from assuming psychotherapeutic roles, particularly in cases where users seek cognitive restructuring from AI systems without professional oversight.

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APPENDIX A
FULL LINEAR PROBE RESULTS

Layer	Pearson	P. Low	P. High	Spearman	Sp. Low	Sp. High	MAE	MAE Low	MAE High
0	0.324044	0.266201	0.382115	0.352267	0.288535	0.413007	0.802567	0.771415	0.837079
1	0.398740	0.342309	0.454767	0.400581	0.341794	0.458066	0.758098	0.722416	0.793970
2	0.487779	0.433412	0.539024	0.497173	0.437734	0.549523	0.703132	0.670294	0.736227
3	0.482537	0.427740	0.530621	0.487575	0.430497	0.539902	0.694587	0.658814	0.730378
4	0.527634	0.473804	0.577025	0.528798	0.476842	0.582676	0.668959	0.633882	0.705664
5	0.522538	0.471566	0.569833	0.530213	0.478463	0.580001	0.679524	0.644633	0.716764
6	0.503915	0.453013	0.547583	0.509057	0.455997	0.564106	0.701977	0.666733	0.738471
7	0.531692	0.484720	0.578807	0.529844	0.472826	0.585189	0.692835	0.657724	0.731350
8	0.552189	0.503468	0.597608	0.551469	0.501120	0.596856	0.682604	0.648046	0.718231
9	0.575967	0.529292	0.620907	0.577873	0.527661	0.624536	0.658237	0.622911	0.693651
10	0.565027	0.519062	0.608080	0.565103	0.514086	0.614572	0.671905	0.636885	0.706574
11	0.577361	0.528111	0.619393	0.578444	0.528421	0.624084	0.659948	0.626229	0.695406
12	0.592202	0.542908	0.632430	0.588639	0.541262	0.634746	0.650037	0.612584	0.683823
13	0.596381	0.549778	0.641160	0.586419	0.528616	0.633541	0.642597	0.607523	0.676702
14	0.631620	0.585176	0.671662	0.622256	0.573650	0.668065	0.613502	0.579467	0.647145
15	0.636194	0.593734	0.677930	0.630530	0.582547	0.672340	0.610545	0.573669	0.643929
16	0.636441	0.593761	0.673465	0.628474	0.578677	0.669102	0.619373	0.584414	0.653617
17	0.614546	0.571677	0.658927	0.614064	0.562783	0.658497	0.635489	0.601062	0.670403
18	0.617652	0.573892	0.657330	0.613480	0.567993	0.659573	0.645422	0.611044	0.681204
19	0.598951	0.552875	0.644982	0.592068	0.542072	0.641359	0.660807	0.626897	0.699008
20	0.611370	0.567619	0.658641	0.604130	0.555858	0.649237	0.650186	0.614355	0.689472
21	0.602919	0.559166	0.647626	0.593613	0.546056	0.638469	0.664524	0.628893	0.699331
22	0.605528	0.560897	0.647008	0.599641	0.545560	0.646027	0.669179	0.638814	0.701560
23	0.593558	0.546166	0.633305	0.588465	0.538116	0.638662	0.675280	0.639257	0.710749
24	0.605556	0.560197	0.646326	0.601903	0.553952	0.647491	0.669003	0.633622	0.704258
25	0.602240	0.557288	0.644134	0.598089	0.546821	0.644395	0.670045	0.637178	0.704098
26	0.605572	0.562625	0.648229	0.603226	0.556476	0.643446	0.667898	0.634102	0.704613
27	0.609078	0.564299	0.648139	0.606081	0.555518	0.650646	0.662226	0.626413	0.697205
28	0.602744	0.555433	0.643074	0.599940	0.552822	0.646238	0.662807	0.629773	0.699621
29	0.613849	0.571907	0.654619	0.609930	0.560270	0.654381	0.654696	0.618886	0.690675
30	0.616019	0.574193	0.656127	0.610428	0.565449	0.655977	0.654548	0.620760	0.690529
31	0.613240	0.566618	0.654089	0.608163	0.561534	0.649875	0.661380	0.625006	0.693766

APPENDIX B
FULL SHUFFLED-PROBE RESULTS

Layer	Pearson	CI Low	CI High	Spearman	CI Low	CI High	MAE	CI Low	CI High
0	0.005781	-0.062625	0.074431	0.010758	-0.055525	0.082727	0.850120	0.814586	0.885837
1	0.043279	-0.025657	0.110996	0.042716	-0.023766	0.113175	0.846198	0.808895	0.884102
2	0.053638	-0.015379	0.116129	0.074312	0.003521	0.144609	0.844337	0.805293	0.881708
3	0.067186	-0.002633	0.134424	0.057272	-0.011082	0.135355	0.856296	0.816642	0.896622
4	0.089900	0.018724	0.166742	0.092062	0.022038	0.165704	0.866069	0.823192	0.908757
5	0.041949	-0.031700	0.115283	0.035884	-0.038410	0.103051	0.915322	0.869685	0.962706
6	0.011768	-0.062877	0.077130	0.013067	-0.056367	0.088683	0.946685	0.902956	0.995305
7	0.034803	-0.029270	0.104717	0.016160	-0.048319	0.089191	0.948788	0.903098	0.996635
8	0.045325	-0.028000	0.118514	0.033931	-0.036293	0.104528	0.934235	0.889844	0.981542
9	-0.022160	-0.089763	0.047898	-0.040970	-0.107519	0.030164	0.974785	0.928167	1.020884
10	-0.010676	-0.082131	0.064655	-0.033688	-0.103004	0.033820	0.976235	0.931095	1.022033
11	0.014867	-0.053345	0.085423	0.014204	-0.061463	0.080899	0.953029	0.909528	1.000710
12	0.023139	-0.047213	0.099183	0.014398	-0.057844	0.081866	0.943315	0.899269	0.988744
13	-0.007972	-0.077466	0.062990	-0.021451	-0.092633	0.047699	0.970582	0.922346	1.019683
14	0.018946	-0.047948	0.091915	0.014708	-0.055946	0.084858	0.959176	0.911831	1.007040
15	0.000438	-0.073646	0.075095	-0.003290	-0.070135	0.076097	0.983540	0.935777	1.031869
16	0.004004	-0.066294	0.073984	-0.005201	-0.077860	0.063069	0.996509	0.946003	1.048471
17	0.026418	-0.037261	0.096907	0.012030	-0.056220	0.083473	1.007120	0.958321	1.060025
18	0.022658	-0.051842	0.088066	0.014683	-0.058464	0.084417	1.020167	0.966062	1.073530
19	0.011183	-0.053441	0.078098	-0.001249	-0.068224	0.067917	1.042266	0.989699	1.092940
20	0.005997	-0.068814	0.077790	-0.009402	-0.080038	0.063234	1.036758	0.978478	1.089571
21	0.006192	-0.064915	0.077612	-0.002844	-0.070344	0.061515	1.032490	0.981511	1.087629
22	-0.000507	-0.069790	0.071722	0.000796	-0.068912	0.070591	1.042519	0.991612	1.097698
23	-0.003858	-0.071331	0.070297	0.000268	-0.069215	0.072448	1.048601	0.996231	1.103178
24	0.030759	-0.039605	0.097148	0.033693	-0.036144	0.103546	1.020114	0.964136	1.075215
25	0.021861	-0.047147	0.089738	0.021873	-0.043239	0.093561	1.039898	0.990787	1.093906
26	0.007562	-0.062781	0.076695	-0.002554	-0.070391	0.067596	1.057474	1.003868	1.110127
27	0.026617	-0.041083	0.096497	0.015671	-0.052156	0.079887	1.040804	0.992305	1.093094
28	0.019732	-0.050550	0.084003	0.003150	-0.071086	0.070637	1.047421	0.998803	1.098727
29	0.043459	-0.024001	0.112323	0.029707	-0.040965	0.098519	1.017408	0.968969	1.066285
30	0.031391	-0.034377	0.099667	0.026884	-0.038805	0.094176	1.010645	0.959589	1.061832
31	0.050945	-0.014026	0.119623	0.042094	-0.028783	0.113836	1.005199	0.957509	1.058705

APPENDIX C
FULL FINAL EVALUATION STEERING RESULTS

Configuration	Mean [95% CI]	SD	Δ [95% CI]	d_z	p_{Holm}	Equiv.
Therapy quality						
Baseline: no steering (α=0)	2.764 [2.702, 2.826]	0.61	—	—	—	—
Steering vector, L6, α=1	2.598 [2.534, 2.662]	0.63	−0.166 [−0.239, −0.091]	−0.23	.002	–
Steering vector, L17, α=1	2.764 [2.708, 2.818]	0.54	+0.000 [−0.070, +0.072]	+0.00	1.000	✓
Random vector, L17, α=1	2.665 [2.601, 2.727]	0.62	−0.099 [−0.174, −0.027]	−0.14	.136	–
Identifies belief						
Baseline: no steering (α=0)	2.338 [2.271, 2.405]	0.66	—	—	—	—
Steering vector, L6, α=1	2.217 [2.150, 2.284]	0.67	−0.121 [−0.201, −0.043]	−0.15	.058	–
Steering vector, L17, α=1	2.373 [2.306, 2.440]	0.67	+0.035 [−0.046, +0.113]	+0.04	1.000	✓
Random vector, L17, α=1	2.375 [2.311, 2.437]	0.61	+0.038 [−0.040, +0.110]	+0.05	1.000	✓
Asks for evidence						
Baseline: no steering (α=0)	2.397 [2.319, 2.472]	0.75	—	—	—	—
Steering vector, L6, α=1	2.330 [2.255, 2.405]	0.74	−0.067 [−0.155, +0.019]	−0.08	1.000	✓
Steering vector, L17, α=1	2.434 [2.367, 2.504]	0.67	+0.038 [−0.056, +0.131]	+0.04	1.000	✓
Random vector, L17, α=1	2.349 [2.273, 2.424]	0.75	−0.048 [−0.142, +0.046]	−0.05	1.000	✓
Avoids validation						
Baseline: no steering (α=0)	3.080 [3.011, 3.147]	0.69	—	—	—	—
Steering vector, L6, α=1	3.088 [3.013, 3.164]	0.73	+0.008 [−0.078, +0.097]	+0.01	1.000	✓
Steering vector, L17, α=1	3.145 [3.072, 3.217]	0.70	+0.064 [−0.024, +0.153]	+0.07	1.000	–
Random vector, L17, α=1	3.080 [3.011, 3.147]	0.68	+0.000 [−0.080, +0.078]	+0.00	1.000	✓
Proposes next step						
Baseline: no steering (α=0)	2.239 [2.150, 2.327]	0.88	—	—	—	—
Steering vector, L6, α=1	2.016 [1.930, 2.102]	0.86	−0.223 [−0.330, −0.110]	−0.21	.003	–
Steering vector, L17, α=1	2.268 [2.182, 2.357]	0.85	+0.029 [−0.070, +0.131]	+0.03	1.000	✓
Random vector, L17, α=1	2.131 [2.043, 2.223]	0.89	−0.107 [−0.209, −0.003]	−0.11	.707	–
Belief flexibility						
Baseline: no steering (α=0)	3.088 [3.046, 3.131]	0.42	—	—	—	—
Steering vector, L6, α=1	3.123 [3.080, 3.166]	0.44	+0.035 [−0.024, +0.094]	+0.06	1.000	✓
Steering vector, L17, α=1	3.105 [3.054, 3.153]	0.48	+0.016 [−0.048, +0.078]	+0.03	1.000	✓
Random vector, L17, α=1	3.099 [3.056, 3.142]	0.42	+0.011 [−0.046, +0.067]	+0.02	1.000	✓
Composite (mean of 6)						
Baseline: no steering (α=0)	2.651 [2.604, 2.697]	0.46	—	—	—	—
Steering vector, L6, α=1	2.562 [2.512, 2.614]	0.50	−0.089 [−0.144, −0.032]	−0.16	.002	–
Steering vector, L17, α=1	2.681 [2.636, 2.727]	0.44	+0.030 [−0.024, +0.084]	+0.06	.281	✓
Random vector, L17, α=1	2.617 [2.571, 2.664]	0.46	−0.034 [−0.088, +0.018]	−0.07	.212	✓
Vector specificity: steering L17 α=1 − random L17 α=1						
Therapy quality	—	—	+0.099 [+0.029, +0.169]	+0.14	.046	–
Identifies belief	—	—	−0.003 [−0.083, +0.078]	−0.00	1.000	✓
Asks for evidence	—	—	+0.086 [−0.005, +0.177]	+0.10	.291	–
Avoids validation	—	—	+0.064 [−0.016, +0.145]	+0.08	.378	✓
Proposes next step	—	—	+0.137 [+0.040, +0.236]	+0.14	.046	–
Belief flexibility	—	—	+0.005 [−0.054, +0.064]	+0.01	1.000	✓
Composite (mean of 6)	—	—	+0.065 [+0.014, +0.115]	+0.13	.014	–

APPENDIX D
SYSTEM LLM PROMPT FOR LLM-AS-JUDGE GRADING

```
You are a rigorous evaluator of CBT-style therapeutic responses for a research
study.
Evaluate each response using the rubric below. Score each category
independently on a 0--4 scale using the operational definitions provided.
Assign the score that best matches the evidence present in the text. Do not
infer intentions that are not explicitly supported by the response.
Do not give partial credit outside the defined scale. Apply the rubric
consistently across all responses.
Scale:
0: Absent or counterproductive. The response contradicts CBT principles,
reinforces maladaptive beliefs, or provides no relevant therapeutic content.
1: Minimal presence. A vague or superficial attempt that lacks clarity,
specificity, or correct CBT structure.
```

2: Partial implementation. Some relevant CBT elements appear but are incomplete, weakly expressed, or inconsistently applied.
3: Adequate implementation. Clear and appropriate use of the CBT element with reasonable specificity and coherence.
4: Strong implementation. Precise, well-structured, and therapeutically appropriate use of the CBT element with clear relevance to the user's statement.
Categories:
therapy_quality: Overall CBT quality of the response. Consider clarity, structure, appropriateness of tone, and whether the response follows a cognitive restructuring framework (acknowledging emotion, identifying cognition, encouraging reflection, and suggesting constructive reframing or action).
identifies_belief: Does the response explicitly identify or restate the underlying belief, interpretation, or automatic thought expressed by the user? High scores require clear recognition of the belief as a thought or interpretation rather than a fact.
asks_evidence: Does the response encourage examination of evidence, alternative explanations, or cognitive flexibility? This may include questions prompting reflection, requests for evidence, or presentation of plausible alternative interpretations.
avoids_validation: Does the response avoid endorsing hopeless, absolute, or distorted beliefs as factual? Appropriate empathy is allowed, but the response should not reinforce cognitive distortions or present them as accurate conclusions.
proposes_next_step: Does the response suggest a concrete, realistic next step consistent with CBT practice? This may include a small behavioral experiment, reflective exercise, journaling prompt, or actionable coping strategy.

APPENDIX E
SYSTEM LLM PROMPT FOR LABELING BELIEF RIGIDITY OF DREADDIT SAMPLES

You are labeling belief rigidity in depressive Reddit text. Return ONLY valid JSON with keys: belief (int 1–5), discard (int 0/1).
Belief scale: 1=highly rigid negative beliefs (absolute, fixed, resistant to change); 5=low rigidity/high flexibility (open to uncertainty, change, alternative interpretations).
Judge the writer's expressed belief structure, not emotional intensity. discard=1 if text is not depressive, irrelevant, incoherent, or too ambiguous to label; otherwise 0. If discard=1, belief should be -1 since it will be ignored. Always return both keys.

APPENDIX F
SYSTEM LLM PROMPT FOR CONTRASTIVE GENERATION

You are generating a balanced cognitive reframe for a single text sample. A balanced reframe acknowledges the difficulty of the situation while avoiding extreme or absolute conclusions. It should introduce a more flexible and realistic interpretation without forcing positivity.
It should be in the same perspective as the original thought, not an outside observer. Act as the person themselves, not a therapist.
Do not shame, invalidate, or dismiss feelings. Keep it concise, natural, and specific. Also decide if the text is viable: discard=1 if the text is not depressive, irrelevant, incoherent, or too ambiguous to label; otherwise 0. If discard=1, the reframe must be an empty string. The text sample could be Nobody at school likes me; I'll always be alone. The corresponding reframe could be I haven't connected with many people at school lately, but I have friends in other circles and I can work on reaching out more.

APPENDIX G
SYSTEM LLM PROMPT FOR PSYCHOTHERAPY REFRAME INFERENCE

```
You are a supportive, non-judgmental assistant helping someone reframe
a difficult thought using principles inspired by cognitive behavioral
therapy. Your response must be written as a single coherent paragraph that is
approximately 80 words. Do not use bullet points, numbered lists, or section
headers.
Begin by briefly acknowledging the emotional experience expressed in the
message. Then gently identify the belief or interpretation the person appears
to be making, framing it as a thought rather than a fact.
Next, encourage balanced reflection by introducing alternative possibilities
or questions that challenge the certainty of the belief without dismissing the
person's feelings." End by offering a realistic and compassionate reframing
that promotes flexibility, hope, or self-understanding.
Avoid diagnosing, giving medical or crisis instructions, or using overly
technical psychological terminology. Avoid overly short responses, repetitive
reassurance, or generic motivational language.
Maintain a calm, respectful, and collaborative tone throughout the paragraph.
Do not include URLs, citations, copyrights, ads, headings, formatting, or any
'service' language.
User:
Assistant:
```

APPENDIX H
SQL QUERIES

A. Query for Probing Dataset

```
SELECT subreddit, text, id
FROM train
WHERE subreddit NOT IN ('food_pantry', 'assistance', 'relationships')
LIMIT 1000;
```

B. Query for CAA Dataset

```
SELECT subreddit, text, id
FROM train
WHERE subreddit NOT IN ('food_pantry', 'assistance', 'relationships')
LIMIT 800 OFFSET 1000;
```